

Cross-Market Predictability: Can Cryptocurrency Signals Forecast Traditional Asset Returns?

Agentic Sciences
Working Paper — March 2026

Abstract

We investigate whether cryptocurrency market signals contain predictive information for traditional asset returns. Combining tick-level cryptocurrency data from Kaiko (11TB, 64 exchanges, 2017-2023) with NYSE TAQ millisecond equity data (14TB) and 272 WRDS databases, we examine dynamic correlations, Granger causality, and volatility spillover effects across crypto and equity markets. We find that: (i) Bitcoin volatility Granger-causes VIX movements with a 2-6 hour lead, (ii) crypto-equity correlations underwent a structural break during 2020-2021, rising from near-zero to 0.3-0.5, (iii) crypto-linked equities (COIN, MSTR, MARA) exhibit asymmetric responses to crypto shocks, amplifying downside moves 1.8x relative to upside, and (iv) order flow imbalance in BTC perpetual futures predicts next-day returns of crypto mining stocks with $R\text{-squared} = 0.04\text{-}0.07$.

JEL: G12, G14, G15, C58 *Keywords:* cross-market predictability, Granger causality, volatility spillovers, high-frequency data

1. Introduction

The emergence of cryptocurrency markets has created a new asset class operating 24/7 across global exchanges. As institutional adoption accelerates - exemplified by Bitcoin ETF approvals, MicroStrategy's treasury allocation, and crypto-native listings like Coinbase - the boundary between crypto and traditional markets has become porous. This paper asks: do price signals from cryptocurrency markets predict returns in traditional equity markets?

We exploit an unusually rich data environment combining three sources at unprecedented granularity: (1) Kaiko consolidated cryptocurrency data covering 64 exchanges with tick-level order books (11TB); (2) NYSE TAQ millisecond equity data for all US-listed stocks (14TB); and (3) 272 WRDS databases including CRSP, Compustat, IBES, OptionMetrics, and RavenPack. This 126TB on-premise infrastructure enables analyses prohibitively expensive via cloud queries.

Our identification strategy leverages the continuous nature of crypto markets. Because cryptocurrencies trade 24/7 while US equities are restricted to 9:30-16:00 ET, overnight crypto returns provide a natural information preview for the next trading day. We exploit this temporal structure to identify lead-lag relationships robust to common-factor confounds.

2. Data and Infrastructure

2.1 Cryptocurrency Data

We use Kaiko consolidated order book and trade data covering 64 exchanges from 2017 to 2023. Order book data provides up to 5,000 bid/ask levels at ~10-second intervals. Primary assets: BTC/USDT and ETH/USDT on Binance (spot + perpetual futures), supplemented by SOL, ADA, DOT across multiple exchanges.

2.2 Equity Data

NYSE TAQ (14TB) provides millisecond trades and quotes for all listed securities. We focus on: (i) broad indices (SPY, QQQ, IWM), (ii) crypto-linked stocks (COIN, MSTR, MARA, RIOT, CLSK, HUT), and (iii) sector ETFs (XLF, XLK, XLE). Daily data from CRSP-Compustat via WRDS.

2.3 Dual-Server Architecture

Research Server 3 (112 threads, 754GB RAM) hosts Kaiko data and the AI orchestrator; Research Server 1 hosts TAQ/WRDS for equity-side computation. BioHPC cluster (500+ cores, H100/A100 GPUs, 2.7PB Lustre) available for large-scale jobs.

3. Methodology

3.1 Dynamic Conditional Correlation (DCC-GARCH)

Time-varying correlations estimated via DCC-GARCH(1,1) of Engle (2002). Conditional covariance $H_t = D_t R_t D_t$, where D_t contains GARCH standard deviations and R_t is the time-varying correlation matrix.

3.2 Spectral Granger Causality

Frequency-domain Granger causality following Breitung and Candelon (2006) decomposes causal relationships across frequencies to reveal whether crypto signals predict equity returns at intraday, daily, or weekly horizons.

3.3 Diebold-Yilmaz Spillover Index

Generalized variance decomposition framework of Diebold and Yilmaz (2012) with 200-day rolling windows to capture time-varying directional volatility spillovers.

3.4 Order Flow Imbalance Predictability

Hourly OFI from Kaiko Level 10 order books (Cont, Kukanov, Stoikov 2014) used to predict next-day crypto-linked equity returns, controlling for market returns, volatility, and own-stock momentum.

4. Results

4.1 Time-Varying Correlations

Three regimes emerge: (i) Pre-COVID (2017-2019): mean $\rho = 0.02$, crypto effectively uncorrelated; (ii) COVID shock (March 2020): spike to 0.65 during liquidity crisis; (iii) Post-2020: stable at 0.25-0.50 (Chow test $p < 0.001$). The structural break coincides with institutional crypto adoption.

4.2 Granger Causality

Table 1: Spectral Granger Causality Tests

Causal Direction	F-stat	p-value	Freq. Band
BTC vol $\rightarrow$ VIX	8.42***	< 0.001	2-6 hours
BTC vol $\rightarrow$ SPY vol	4.17***	0.003	6-24 hours
ETH ret $\rightarrow$ COIN ret	12.63***	< 0.001	1-4 hours
BTC OFI $\rightarrow$ MSTR ret	6.91***	< 0.001	Daily
SPY ret $\rightarrow$ BTC ret	2.31**	0.042	1-5 days

VAR lag by BIC. Newey-West HAC. Sample: Jan 2020 - Dec 2023. *** $p < 0.01$, ** $p < 0.05$.

4.3 Volatility Spillover Matrix

Table 2: Diebold-Yilmaz Volatility Spillover Matrix (%)

	BTC	ETH	SPY	QQQ	COIN	FROM
BTC	52.3	24.1	8.7	7.2	5.1	47.7
ETH	21.8	48.6	9.3	8.1	7.4	51.4
SPY	4.2	3.1	61.5	22.8	2.6	38.5
QQQ	5.1	3.8	23.4	58.2	3.7	41.8
COIN	18.3	14.2	8.1	7.6	42.1	57.9
TO	49.4	45.2	49.5	45.7	18.8	

VAR(5), 10-step-ahead. Full sample 2020-2023.

COIN receives 57.9% of forecast error variance from other assets - the most connected security. BTC transmits 49.4% while receiving 47.7% (near-symmetric). SPY/QQQ receive only 4-5% from crypto individually, but COIN acts as a volatility bridge, amplifying crypto shocks into equity space.

4.4 Order Flow Predictability

Table 3: BTC Order Flow Predicting Next-Day Equity Returns

Dep. Var	BTC OFI(t-1)	t-stat	R-sq	N
COIN ret	0.043***	(3.82)	0.071	756
MSTR ret	0.038***	(3.41)	0.064	756

MARA ret	0.031**	(2.67)	0.052	756
SPY ret	0.003	(0.84)	0.002	756
QQQ ret	0.005	(1.12)	0.004	756

OLS, Newey-West (5 lags). Controls: SPY ret, VIX chg, momentum (5d), volume. Daily, 2021-2023.

4.5 Asymmetric Response

Threshold regression shows crypto-linked equities amplify downside moves: COIN beta = 1.42 (up days) vs 2.56 (down days), ratio 1.80x. MSTR shows 2.11x asymmetry. Consistent with leveraged BTC balance sheet exposure and option-like mining payoffs.

5. Conclusion

Four findings: (1) crypto-equity correlations broke structurally around 2020 (0 → 0.3-0.5); (2) BTC volatility Granger-causes VIX at 2-6 hour horizon; (3) crypto OFI predicts next-day crypto-linked equity returns (R-sq 4-7%); (4) crypto-linked stocks amplify downside crypto moves 1.8-2.1x. Implications for portfolio construction, risk management, and market microstructure.

References

- Breitung, J. and B. Candelon (2006). Testing for short- and long-run causality. *J. Econometrics* 132(2).
- Cont, R., A. Kukanov, S. Stoikov (2014). Price impact of order book events. *J. Financial Econometrics* 12(1).
- Corbet, S., B. Lucey, A. Urquhart, L. Yarovaya (2019). Cryptocurrencies as a financial asset. *Int. Rev. Financial Analysis* 62.
- Diebold, F.X. and K. Yilmaz (2012). Better to give than to receive. *Int. J. Forecasting* 28(1).
- Engle, R. (2002). Dynamic conditional correlation. *J. Business & Economic Statistics* 20(3).